MILITARY SPECIFICATION

LENS ASSEMBLY, EYEPIECE, FOR NIGHT VISION GOGGLES, AN/PVS-5( )

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The specification covers the Lens Assembly, Eyepiece, referred to herein as the eyepiece, which is part of Night Vision Goggles, AN/PVS-5( ).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	- Pallet, Materials Handling, Wood, Stringer Construction, 2-Way and 4-Way (partial)
QQ-S-781	- Strapping, Steel, Flat and Seals
NNN-P-40	- Paper, Lens
PPP-B-566	- Boxes, Folding, Paperboard
PPP-B-585	- Boxes, Wood, Wirebound
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock Corner
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple Wall
PPP-C-843	- Cushioning Material, Cellulosic
PPP-S-760	- Strapping, Nonmetallic, (and Connectors)
PPP-T-97	- Tape, Packaging/Industrial, Filament Reinforced

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of
MIL-B-117	- Bags, Sleeves and Tubing-Interior Packaging
MIL-C-675	- Coating of Glass Optical Elements
MIL-O-13830	- Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Cdr, USACERCOM, ATTN: DRSEL-RD-TS-S, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-150	- Photographic Lenses
MIL-STD-171	- Finishing of Metal and Wood Surfaces
MIL-STD-454	- Standard General Requirement for Electronic Equipment
MIL-STD-810	- Environmental Test Methods
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment

DRAWINGS

SM-D-804195	- Cell Assembly Eyepiece

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Construction. The Eyepiece shall be in accordance with SM-D-804195 and as specified herein.

3.2 First article. Supplier shall furnish and test eyepieces as specified in the contract or purchase order (see 6.2).

3.3 Parts and materials. Parts and materials shall be as specified herein and as shown on applicable drawings. Materials not specified shall be selected by the contractor.

3.4 Performance characteristics.

3.4.1 Equivalent focal length. The equivalent focal length of the eyepiece shall be 27.0 mm, ±0.3 mm.

3.4.2 Flange focal distance. The flange focal distance of one eyepiece cell assembly shall be 0.410 inch, ±0.020 inch, where the measurement is taken from the side of the 1.5490-inch diameter closest to the focal plane.

3.4.3 Concentricity. The image displacement of the mechanical and optical axes of the eyepiece shall not exceed 0.005 inch as shown on SM-D-804195.

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3.4.4 Linear distortion. The linear distortion of the eyepiece at the edge of the 18 mm format shall be between 7.0 and 9.5 percent pincushion distortion.

3.4.5 Field of view. The field of view of the eyepiece shall be 40°, ±2° for the 18 mm format.

3.4.6 Exit pupil vs eye relief. The exit pupil diameter shall be no less than 9.8 mm at an eye relief of 15 mm, ±0.2 mm for zero diopter setting.

3.4.7 Field flatness/field tilt. The field flatness/field tilt shall deviate no more than ±1/3 diopter over an 18 mm format with a 40 mm concave radius of curvature at the zero-diopter image plane of the eyepiece.

3.4.8 Transmission. The transmission of the eyepiece shall be not less than 78 percent over the full eyepiece aperture for the spectral output of the P-20 phosphor.

3.4.9 Modulation transfer function (MTF). The MTF of the eyepiece shall be not less than that specified in Figures 1 and 2.

3.4.10 Torque. The eyepiece lens assembly shall withstand a minimum torque of 10 foot-pounds (ft-lbs) applied to the 1.5490 inch diameter flange while the optical cell (see 6.3.5) is held stationary without damage or relative movement.

3.4.11 Weight. The weight of the eyepiece shall not exceed 65 grams.

3.4.12 Design verification. The off axis MTF of the eyepiece vs pupil decentration shall not be less than that specified in Figures 3, 4, 5, and 6.

3.5 Environmental.

3.5.1 Vibration. The eyepiece shall not be damaged (see 6.3.1) after vibration in the frequency range of 10 to 55 hertz at an amplitude level of 0.05 inch ± 0.005 inch double amplitude.

3.5.2 Temperature, humidity, altitude. The eyepiece shall not be damaged during storage at temperatures ranging from -57°C to +65°C, altitudes ranging from sea level to 50,000 feet above sea level while at a temperature of -57°C, a humidity level up to 95 percent relative humidity at a temperature of +65°C.

3.5.3 Temperature shock. The eyepiece shall not be damaged during temperature changes from -57°C to +65°C in 5 minutes or less and from +65°C to -57°C in 5 minutes or less.

3.5.4 Immersion. The eyepiece shall not be damaged and shall not show any evidence of water or moisture within the lens or leakage between the optical cell and any sleeve which may be affixed to it in fabricating SM-D-804195 after being immersed in fresh water to a depth of not less than 3 feet for a period of not less than 30 minutes.

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3.6 Treatment and painting. Unless otherwise specified, the metallic parts of the eyepiece shall be given a treatment and painting per MIL-STD-171 and optical surfaces shall be coated in accordance with MIL-C-675.

3.7 Marking. The eyepiece shall be marked in accordance with SM-D-804195.

3.8 Workmanship. All eyepieces shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristics. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing. Eyepieces shall be free from burrs and shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material. The cleaning method shall not cause damage to any of the eyepieces, nor shall the parts be contaminated by the cleaning agent. Scratches and digs shall be as specified in the optical glass surface quality requirements and the cement defects of MIL-O-13830.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (4.3).
b. Quality conformance inspection (4.4).
c. Inspection for packaging (4.7).

4.3 First article inspection. Twenty-four eyepieces manufactured with production tooling shall be provided for first article and each shall be inspected in accordance with Tables I and II, subgroup I. In addition two of these eyepieces selected at random shall be subjected to first article inspections in Table II, subgroup II. The tests may be performed in any order except that the Modulation Transfer Function shall be performed last. Failure of any test of Table I or Table II may be cause for rejection of first article. If molding is used for manufacture of optical elements, optical cells, or any part thereof, the first article samples shall be chosen so as to have equal representation from each and all molds.

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TABLE I. Inspections.

Categories	Defect	Method of Inspection
See SM-D-804195		
Critical	None defined	
Major		
101	Location of the three pins not in true position	Gage
102	1.5490 diameter out of tolerance	Gage
103	1.2430 diameter out of tolerance	Gage
104	Concentricity between 1.5490 diameter and 1.2430 diameter out of tolerance	
105	1.125 cell diameter out of tolerance	Gage
106	0.590 shoulder length out of tolerance	Gage
Minor		
201	0.100 flange thickness out of tolerance	Gage
202	Marking, missing or not as specified	Visual
203	0.020 by 35° chamfer missing	
204	Surface finish of 1.243 diameter surface not as specified	Visual/Gage
205	Packaging, missing or not as specified	Visual
206	Workmanship (see para 3.8)	Visual

TABLE II. First article inspection.

Test	Requirement paragraph	Test paragraph
Subgroup I		
Equivalent focal length	3.4.1	4.5.1
Flange focal distance	3.4.2	4.5.2
Concentricity	3.4.3	4.5.3
Field of view	3.4.5	4.5.5
Exit pupil vs eye relief	3.4.6	4.5.6
Transmission	3.4.8	4.5.8
Torque	3.4.10	4.5.10
Weight	3.4.11	4.5.11
Vibration	3.5.1	4.6.1
Temperature, humidity, altitude	3.5.2	4.6.2
Temperature shock	3.5.3	4.6.3
Immersion	3.5.4	4.6.4
Modulation transfer function	3.4.9	4.5.9

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TABLE II. First article inspection. (Cont)

Test	Requirement paragraph	Test paragraph
Subgroup II		
Linear distortion	3.4.4	4.5.4
Field flatness/field tilt	3.4.7	4.5.7
Design verification	3.4.12	4.5.12

4.4 Quality conformance inspection.

4.4.1 Inspections. Eyepieces shall be inspected for the attributes listed in Table I. Sampling for major defects shall be per MIL-STD-105, Level II, AQL 1 percent. Sampling for minor defects shall be per MIL-STD-105, Level II, AQL 4 percent.

4.4.2 Tests. For purposes of Group A, Group B, and Group C testing a lot shall be defined as 50 eyepieces or fraction thereof, or one month production, whichever occurs first.

4.4.2.1 Group A tests. Ten eyepieces from each lot shall be randomly selected and tested for the attributes listed in Table III. Failure of an eyepiece to meet the requirements specified shall be cause for rejection of that eyepiece. Failure of 2 or more eyepieces, from the same lot, to meet the requirements specified shall be cause to revert to 100 percent inspection of the lot.

TABLE III. Group A tests.

Test	Requirement paragraph	Test paragraph
Flange focal distance	3.4.2	4.5.2
Concentricity	3.4.3	4.5.3

4.4.2.2 Group B tests. Lots which have passed the inspections of 4.4.2.1 shall be inspected in accordance with Table IV. Sampling for inspection shall be as described herein.

4.4.2.2.1 Sampling for Group B, subgroup I. One eyepiece from each lot that has passed Group A tests shall be randomly selected and tested for the Group B, subgroup I attributes listed in Table IV. Failure to meet the requirements specified shall be cause for rejection of that eyepiece and cause to revert to 100 percent inspection of the lot for the failed attribute until such time as 5 consecutive eyepieces have passed Group B, subgroup I tests.

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4.4.2.2.2 Sampling for Group B, subgroup II. One eyepiece shall be randomly selected from every third lot that has passed Group B, subgroup I inspections and tested for the Group B, subgroup II attributes listed in Table IV. Failure to meet the requirements specified shall be cause for rejection of that eyepiece and cause to revert to 100 percent inspection of the lot for the failed attribute until such time as 5 consecutive eyepieces have passed Group B, subgroup II inspections.

TABLE IV. Group B tests.

Test	Requirement paragraph	Test paragraph
Subgroup I		
Equivalent focal length	3.4.1	4.5.1
Transmission	3.4.8	4.5.8
Modulation transfer function	3.4.9	4.5.9
Torque	3.4.10	4.5.10
Subgroup II		
Linear distortion	3.4.4	4.5.4
Field of view	3.4.5	4.5.5
Exit pupil vs eye relief	3.4.6	4.5.6
Field flatness/field tilt	3.4.7	4.5.7
Weight	3.4.11	4.5.11

4.4.2.3 Group C tests. Lots which have passed the inspection of 4.4.2.2 shall be inspected in accordance with Table V. Sampling for inspection shall be as described herein.

4.4.2.3.1 Sampling for Group C, subgroup I. One eyepiece shall be randomly selected from each lot of eyepieces produced and subjected to the inspections of Table V, subgroup I. Failure of any eyepiece shall be cause for failure of Group C, subgroup I.

4.4.2.3.2 Sampling for Group C, subgroup II. One eyepiece shall be randomly selected from each lot produced during alternate months and subjected to the inspections of Table V, Subgroup II. Testing shall be performed in the order listed in Table V. The first samples selected shall be from the second month production with subsequent samples selected every second month thereafter. Failure of any eyepiece shall be cause for failure of Group C, subgroup II.

TABLE V. Group C inspections.

Test	Requirement paragraph	Test paragraph
Subgroup I		
Immersion	3.5.4	4.6.4

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TABLE V Group C inspections. (Cont)

Test	Requirement paragraph	Test paragraph
Subgroup II		
Vibration	3.5.1	4.6.1
Temperature, humidity, altitude	3.5.2	4.6.2
Temperature shock	3.5.3	4.6.3

4.4.2.4 Actions required in the event of Group C failure. In the event of failure of the inspections of Group C, Subgroup I, all eyepieces represented by the samples shall be subjected to the tests of Subgroup I. Failure of any eyepiece to pass the inspection shall be cause to reject that eyepiece. In the event of failure of the inspections of Group C, Subgroup II, actions as specified in the contract shall be taken (see 6.2).

4.5 Test procedure. Unless otherwise specified all tests shall be performed at room temperature. Temperature chambers shall maintain specified temperatures within $\pm 1.4°C$. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment.

4.5.1 Equivalent focal length. The equivalent focal length of the eyepiece shall be measured in accordance with MIL-STD-150. Failure to meet the requirements of 3.4.1 shall constitute failure of this test.

4.5.2 Flange focal distance. The flange focal distance of the eyepiece shall be measured in accordance with the procedures outlined in MIL-STD-150. Failure to meet the requirements of 3.4.2 shall constitute failure of this test.

4.5.3 Concentricity. Concentricity shall be measured in accordance with MIL-STD-150. Failure to meet the requirements of 3.4.3 shall constitute failure of this test.

4.5.4 Linear distortion. The linear distortion of the eyepiece shall be measured using procedures outlined in MIL-STD-150. In order to use these procedures, the eyepiece shall be run backwards (i.e., collimated light enters the eye/lens) with a 5 mm minimum exit pupil at a 15 mm minimum eye relief distance between the collimator and the eyepiece. Failure to meet the requirements of 3.4.4 shall constitute failure of this test.

4.5.5 Field of view. Field of view of the eyepiece shall be calculated using the data obtained for 9 mm semi-field height in the distortion test 4.5.4. Failure to meet the requirements of 3.4.5 shall constitute failure of this test.

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4.5.6 Exit pupil vs eye relief. Using a diopterscope, center an 18 mm, +0.1 mm -0.0 mm back-illuminated target at the infinity focal plane. Position an opaque mask at 15 mm, ±0.2 mm eye relief distance. The mask shall have two pin holes (1 mm maximum diameter) 9 mm apart (center-to-center) and centered about the optical axis. View through each pin hole to determine if the entire 18 mm target is visible through each pin hole. The opaque mask may not be moved between viewings. Failure to observe the entire 18 mm target through both pin holes shall constitute failure of this test.

4.5.7 Field flatness/field tilt. Field flatness/field tilt of the eyepiece shall be measured in accordance with MIL-STD-150. In order to use this procedure, the eyepiece shall be run backwards (i.e., collimated light enters the eye/lens) with a 5 mm, ±0.2 mm exit pupil at a 15 mm, ±0.2 mm eye relief distance between the collimator and the eyepiece. Failure to meet the requirements of 3.4.7 shall constitute failure of this test.

4.5.8 Transmission. Transmission of the eyepiece shall be measured in accordance with MIL-STD-150. Failure to meet the requirements of 3.4.8 shall constitute failure of this test.

4.5.9 Modulation transfer function. A sine wave pattern generator or equivalent with an S-11 photomultiplier tube shall be used. A green filter (Wratten No. 58 or equivalent over a 2856 K, ±200 K tungsten light source) shall simulate P-20 illumination in the test set-up. If a microscope objective is used to relay the image of the lens, the microscope objective numerical aperture must be greater than or equal to 0.28 (with a minimum magnification of 10X). The pattern-collimator combination must allow the lens to be tested either continuously or at a minimum of six distinct, evenly distributed points from 0 lp/mm to 40 lp/mm with the first point being less than or equal to 1 referenced to the focal plane of the eyepiece.

4.5.9.1 MTF, on-axis, centered and off-centered pupil. Place a 5 mm, ±0.2 mm stop at the 15 mm, +0.0 mm -0.2 mm eye relief distance from the eyepiece. Center this pupil on the optical axis of the eyepiece. Measure on-axis MTF. Off-center this pupil 1 mm, 2 mm, 4 mm, and 5 mm in a direction perpendicular to the eyepiece optical axis and measure on-axis MTF at each pupil position. Failure to meet the requirements of 3.4.9 shall constitute failure of this test.

4.5.9.2 MTF, off-axis, centered pupil. Place a 5 mm ±0.2 mm stop at the 15 mm, +0.0mm -0.2 mm eye relief distance from the eyepiece. Center this pupil on the optical axis of the eyepiece. Measure the off-axis, radial, MTF at a semi-field angle of 14°. Measure the off-axis, tangential, MTF at a semi-field angle of 14°. Failure to meet the requirements of 3.4.9 shall constitute failure of this test.

4.5.10 Torque. While holding the optical cell toward the eye lens end apply a minimum torque of 10 foot pounds (ft-lbs) about the optical axis to the largest diameter flange in a clockwise direction, then in a counter clockwise direction. Repeat this two additional times. Failure to meet the requirements of 3.4.10 shall constitute failure of this test.

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4.5.11 Weight. The eyepiece shall be placed on a suitable scale. Failure to meet the requirements of 3.4.11 shall constitute failure of this test.

4.5.12 Design verification. Place a 5 mm, ±0.2 mm stop at the 15 mm, +0.0 mm -0.2 mm eye relief distance from the eyepiece. Off center this pupil in magnitude and direction as shown in Figures 3, 4, 5, and 6 and measure off axis MTF at each pupil position. Failure to meet the requirements of 3.4.12 shall constitute failure of this test.

4.6 Environmental. Each eyepiece shall demonstrate its ability to perform properly after being subjected to all the Group C, Subgroup II tests. For the purpose of post environmental testing, "perform properly" shall be defined as the ability to successfully complete, in order, the torque (4.5.10), modulation transfer function (4.5.9), transmission (4.5.8), and flange focal distance (4.5.2) tests.

4.6.1 Vibration. The eyepiece shall be mounted to a test fixture and subjected to harmonic motion applied in each of three orthogonal planes; the frequency of the harmonic motion shall be varied uniformly from 10 to 55 Hz and return to 10 Hz in one minutes; the amplitude shall be 0.050 inch, ±0.005 inch double amplitude; this frequency sweep shall be repeated 10 times in each plane. Failure to meet the requirements of 3.5.1 shall constitute failure of this test.

4.6.2 Temperature, humidity, altitude. The eyepiece shall be tested in accordance with method 518.1 of MIL-STD-810 except that in steps 3 and 13 the temperature shall be -57°C and the eyepiece will be inspected for the presence of internal moisture, fogging, or frosting at the completion of steps 3, 5, 7, 9, 13, and 14. Step 15 shall be limited to a visual check for damage. Lens caps may be used on the eyepieces during test. Failure to meet the requirements of 3.5.2 shall constitute failure of this test.

4.6.3 Temperature shock. The eyepiece shall be tested in accordance with method 503.1 of MIL-STD-810 except that the high temperature extreme shall be +65°C and the eyepiece shall be held at the temperature extremes for 2 hours in each step instead of 4 hours. Step 9 shall be limited to a visual check for damage. Failure to meet the requirements of 3.5.3 shall constitute failure of this test.

4.6.4 Immersion. The eyepiece shall be subjected to Test Method 512.1, Procedure I of MIL-STD-810 except that the immersion time shall be 30 minutes minimum. Failure to meet the requirements of 3.5.4 shall constitute failure of the first half of the test. The eyepiece shall then be placed in a fixture such that the object end is inside a water tight container which seals around the 1.243-inch diameter shoulder of the eyepiece and the eye lens end is exposed to the environment and subjected to Test Method 512.1, Procedure I of MIL-STD-810 except that the immersion time shall be 30 minutes minimum. Failure to meet the requirements of 3.5.4 shall constitute failure of the test.

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4.7 Inspection for packaging. Preparation for packaging shall be inspected in accordance with MIL-P-116 to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Eyepieces shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Eyepieces shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical leterature shall be unit packed method IC-1.

5.1.1.4.2 Lens assembly, eyepiece. Each eyepiece shall be individually unit packed method IA-8 as follows: wrap each unit with lens tissue conforming to NNN-P-40. Cushion each eyepiece by wrapping with cushioning material conforming to PPP-C-843, type II. Place the cushioned item within a barrier bag fabricated of material conforming to MIL-B-117, type II, class E, style 1. Heat seal the closure, leaving a small opening for exhausting air. Exhaust the entrapped air and complete the seal. Place the bagged item within a close-fitting box conforming to PPP-B-566, variety 2. Place the technical literature unit packed as specified in 5.1.1.4.1 on top of the contents directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.1.4.3 Intermediate packing. Multiples of 5 not to exceed 20 eyepieces unit packed as specified in 5.1 shall be placed within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservation application, unit packing and intermediate packing shall be as specified in 5.1, except that method III shall be used in liue of method IA-8.

5.1.3 Commercial preservation. Preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

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5.2.1 Level A. A quantity of eyepieces, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.2 Level B. A quantity of eyepieces, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The eyepiece is used with the AN/PVS-5( ) for general purpose viewing during period of darkness.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Time frame required for submission of first article and number of assemblies required (see 3.2).
c. Disposition of Group B assemblies.
d. Level A, level B, or Commercial preservation and packing (see section 5).
e. Necessary action by contractor in event of Group B failures.
f. When rough handling test is required.

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g. Environmental pollution prevention measures are contained in the packaging material specifications reference herein. Refer to material specification or preparing activity for recommended disposability methods.

h. To guard against the use of material with hazardous levels of radioactive material data item DI-T-1903, DI-T-1906, and supplement(s) thereto shall be a part of any contract incorporating this specification.

i. For eyepieces being procured by the Government as an integral part of Night Vision Goggles, AN/PVS-5( ) the inspections of Table I, The Group A tests, the weight test and the environmental testing per this specification shall be waived.

6.3 Definitions.

6.3.1 Damage. Damage is defined as failure of any finish, condensation, fogging or moisture on any internal surface; dents, or breakage of any hardware, connection, or component.

6.3.2 Inspection lot size. An inspection lot is defined as not more than those assemblies successfully completing Group A tests each month.

6.3.3 Environmental tests. For the purpose of this specification, environmental tests are those tests discussed in 4.6.

6.3.4 Room temperature. Room temperature is defined as +23°C, ±10°C.

6.3.5 Optical cell. The optical cell is defined as the cylindrical part which is in direct contact with the lens elements, holds the lens elements in their fixed relative position, and has no external sleeves, flanges, or other surfaces or fittings attached to it in any manner.

Custodian:
Army - ER
Air Force - 11

Review activity:
Army - AT
Air Force - 99

User activity:
Army - AT

Preparing activity:
Army - ER

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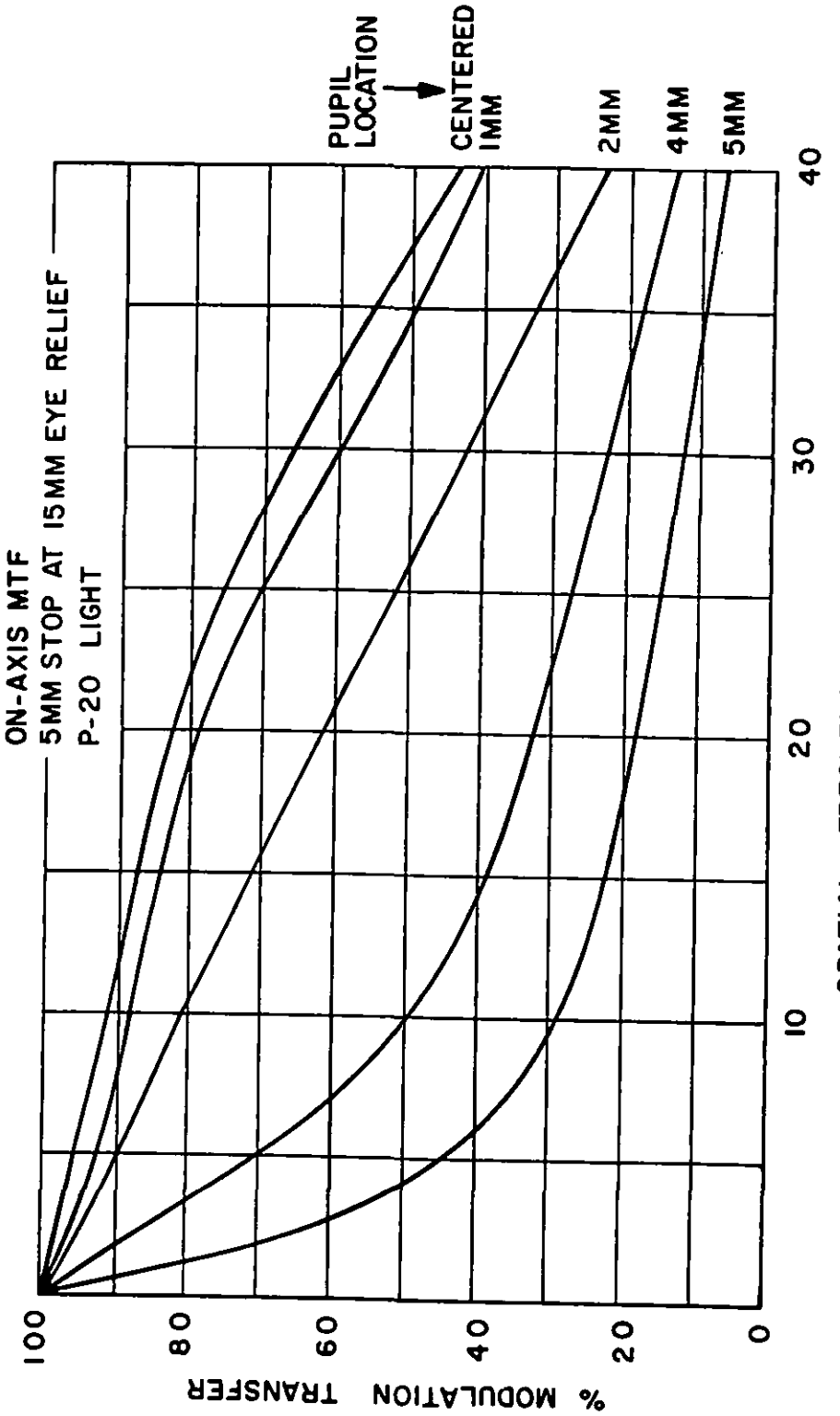


FIGURE 1. ON-AXIS MTF, EYEPIECE

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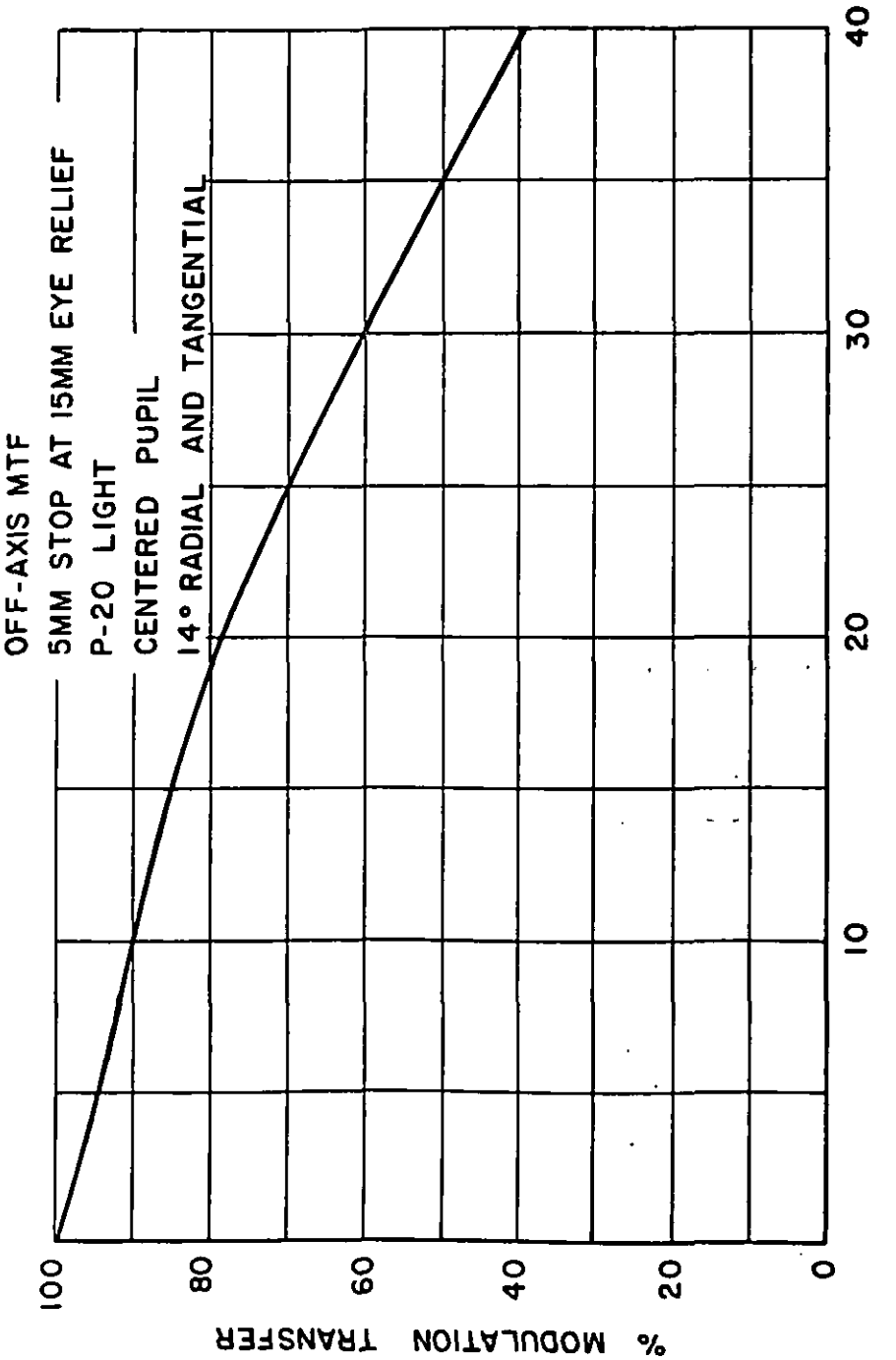


FIGURE 2. OFF-AXIS MTF, EYEPIECE

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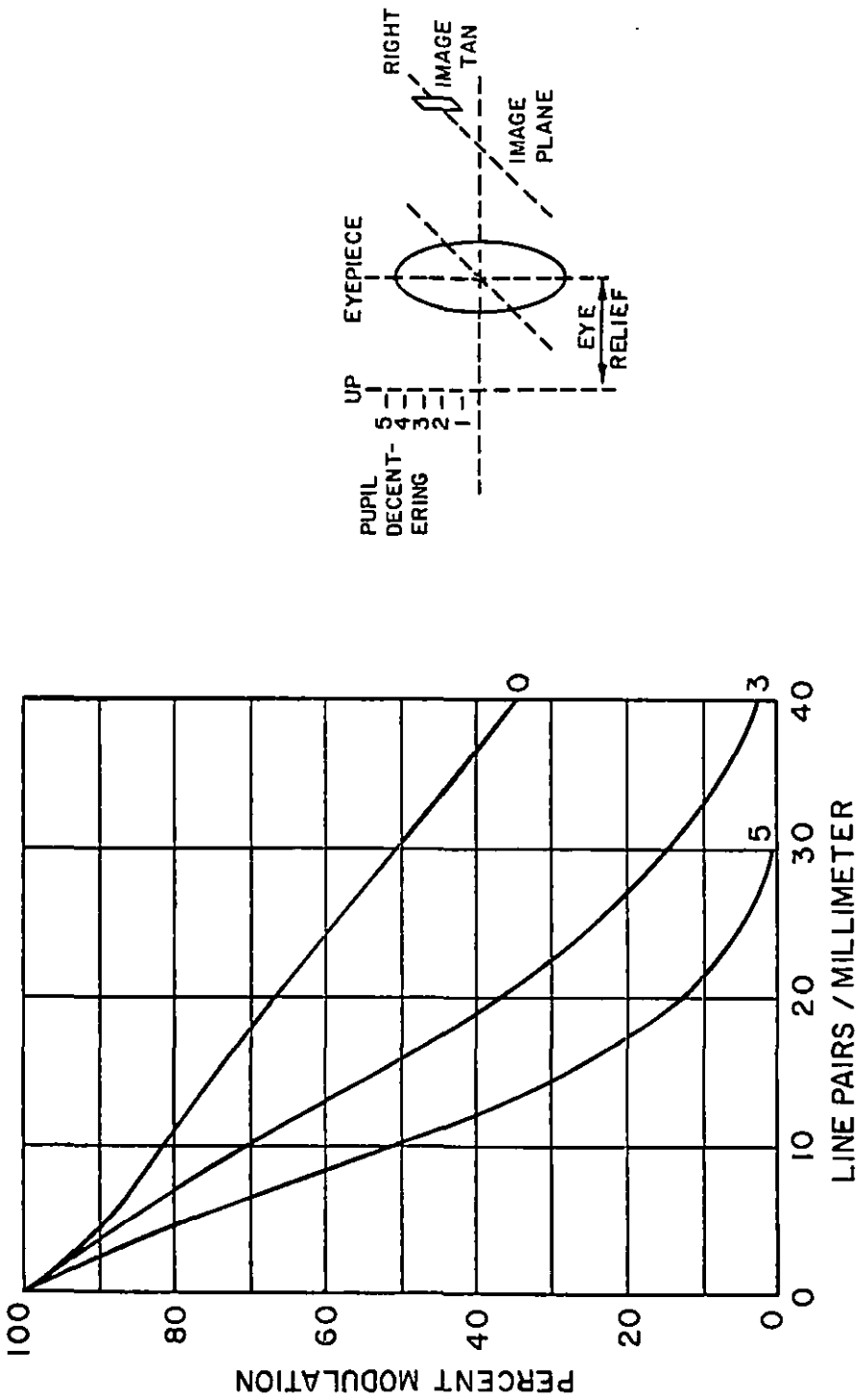


FIGURE 3. OFF-AXIS TANGENTIAL MTF WITH PARALLEL PUPIL DECENTRATION

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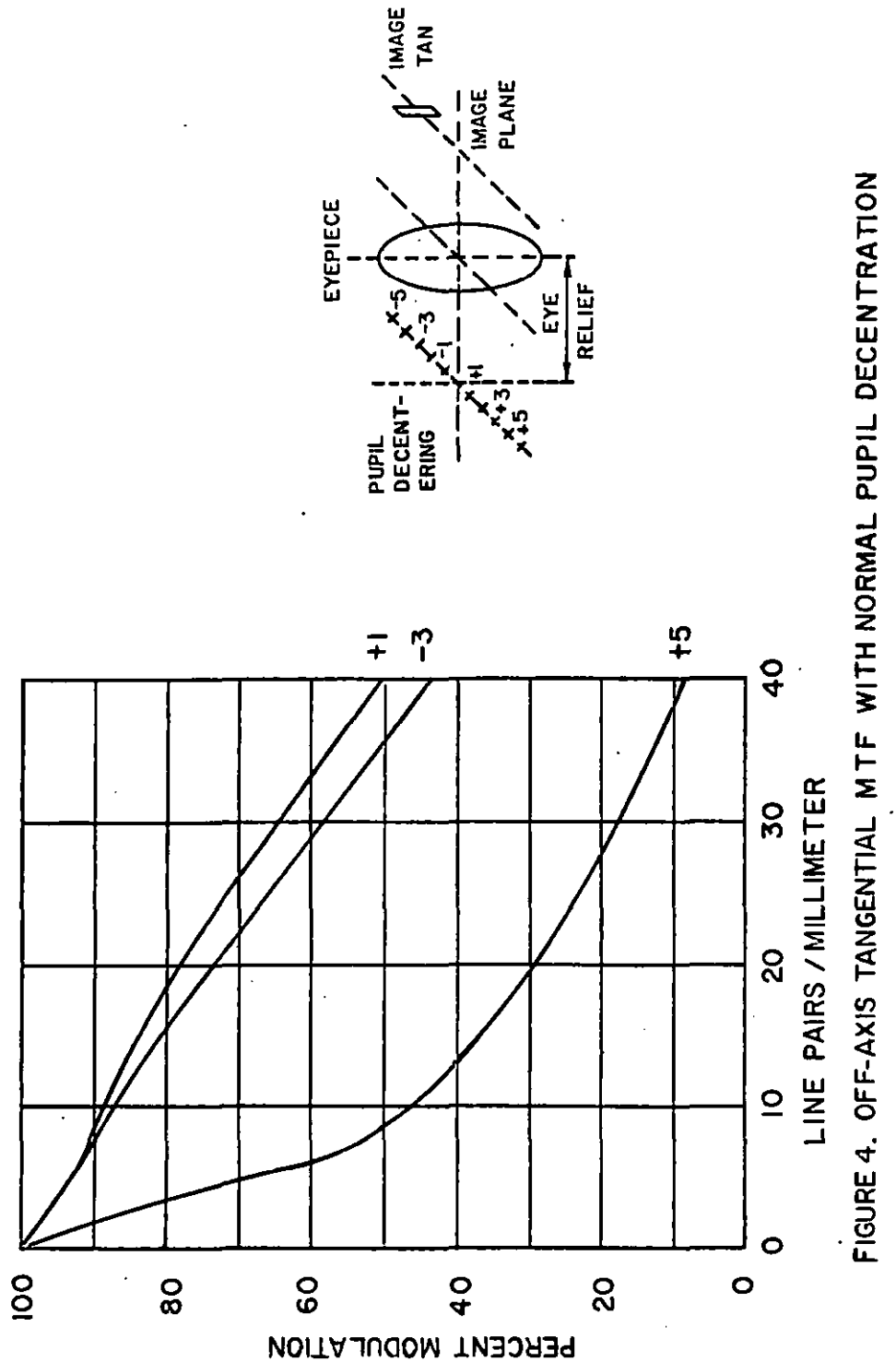


FIGURE 4. OFF-AXIS TANGENTIAL MTF WITH NORMAL PUPIL DECENTRATION

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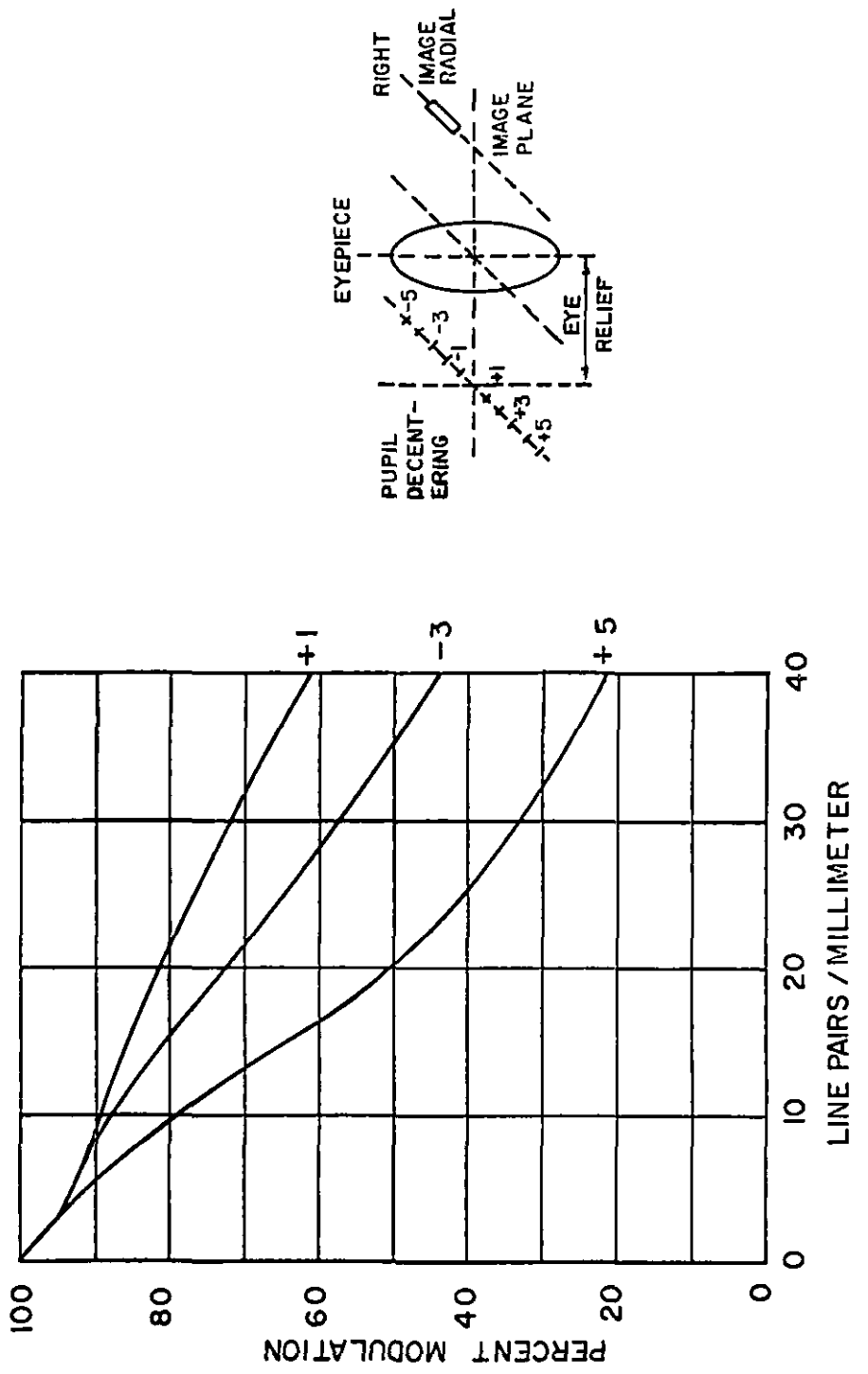


FIGURE 5. OFF-AXIS RADIAL MTF WITH PARALLEL PUPIL DECENTRATION

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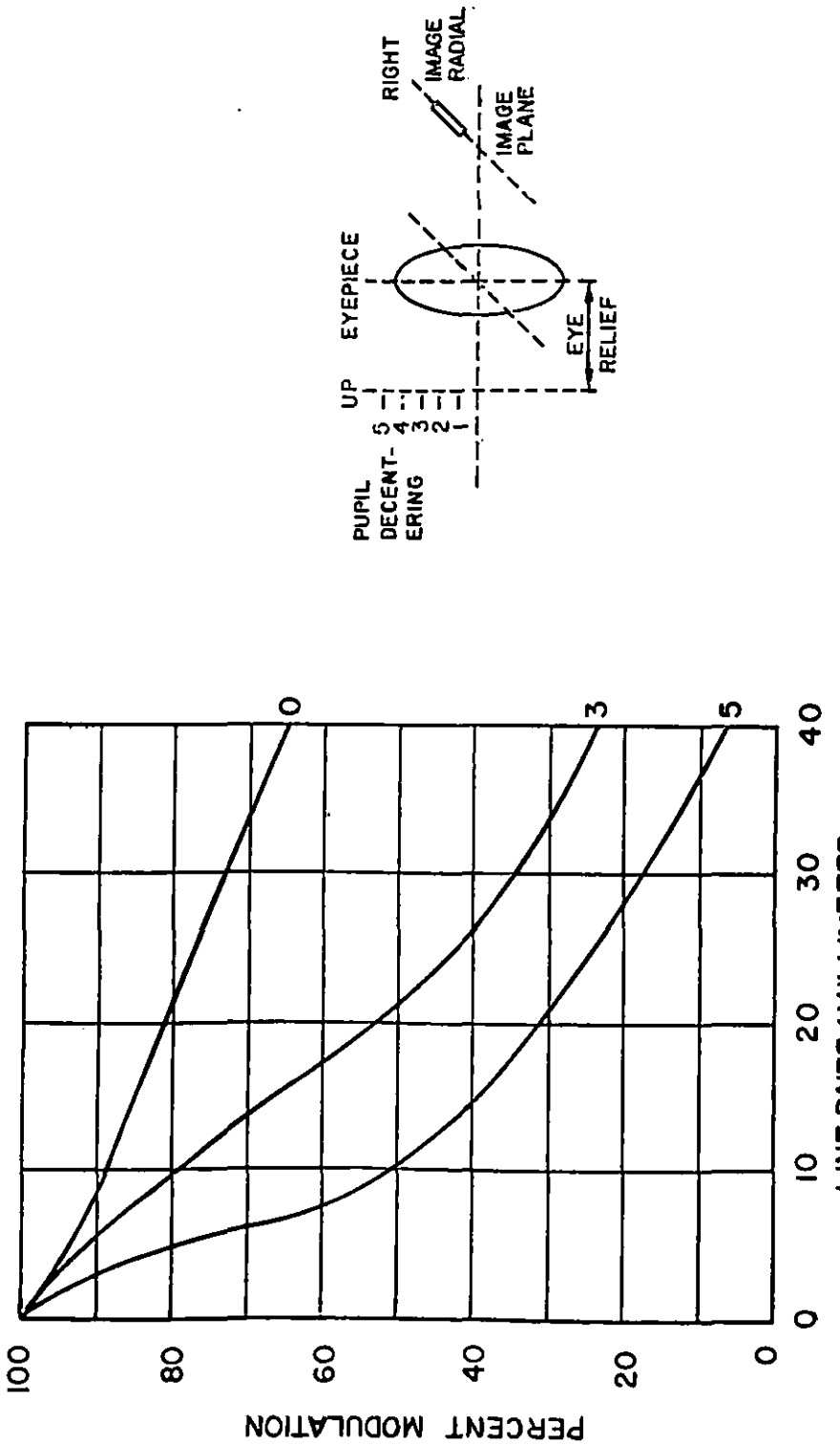


FIGURE 6. OFF-AXIS RADIAL MTF WITH NORMAL PUPIL DECENTRATION

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